L1

Linear Systems

08th Sept 2020

Transfer function

$$G(s) = Y(s)/U(s)$$

Laplace transform

Impulse response

$$y(t) = \underbrace{h(t)}_{\text{Impulse response}} * u(t)$$

$$G(s) = C(sI-A)^{-1}B + D$$

First order rep^n

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$
$$A \in \mathbb{R}^{n\times n},\ B \in \mathbb{R}^{n\times m}$$
$$C \in \mathbb{R}^{p\times n},\ D \in \mathbb{R}^{p\times p}$$

$$x(t) = e^{At}x_0 + \underbrace{\int_0^t e^{A(t-\tau)}Bu(\tau)d\tau}_{\text{Convolution}}$$

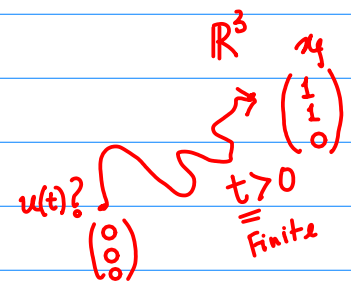
$$\Updownarrow$$

$$\begin{bmatrix} \dot{x}(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \longrightarrow \text{Compact form of writing the system}$$

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] := C(sI-A)^{-1}B + D$$

Reachability (Controllability)

Given a vector $x_f \in \mathbb{R}^n$ is it possible to steer the states $x(t)$ from 0 to x_f in finite time t_f with piecewise-continuous input $u(t)$?



$$\dot{x} = Ax + Bu \Rightarrow x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau$$

$x_0 = 0$

$$x(t) = \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau \longrightarrow (1)$$

$$\boxed{e^{At} = I + At + \frac{At^2}{\lfloor 2} + \cdots}$$

Cayley-Hamilton Th^m: A matrix satisfies its own characteristic equation.

(A) $a_0 + a_1 s + \cdots a_{n-1}s^{n-1} + s^n = 0 \quad a_i \in \mathbb{R} \Rightarrow a_0 I + a_1 A + \cdots + a_{n-1}A^{n-1} + A^n = 0$

$A \in \mathbb{R}^{n\times n}$

Characteristic eq^n $\rightarrow \det(sI-A) = 0$

$\downarrow$

n-deg polynomial

$$a_0 I + a_1 A + a_2 A^2 + \dots a_{n-1} A^{n-1} + A^n = 0$$

$$\Rightarrow \quad A^n = -a_0 I - a_1 A - a_2 A^2 - \dots - a_{n-1} A^{n-1}$$

$$\Rightarrow \quad A^n = \text{linear comb}^n \left(I, A, A^2 \dots A^{n-1}\right)$$

If $k > n-1$. Then $\exists$ scalar constants $\alpha_0, \alpha_1 \dots \alpha_{n-1}$ s.t.

$$A^k = \alpha_0 I + \alpha_1 A + \dots \alpha_{n-1} A^{n-1}$$

$$A^k \in \text{span}\{I, A, \dots A^{n-1}\}$$

$$e^{At} = I + At + A^2 t^2/12 + \dots\dots = \phi_0(t) I + \phi_1(t) A + \dots + \phi_{n-1}(t) A^{n-1}$$

$$\forall\, t \geq 0$$

$$① \Rightarrow \quad x(t) = \int_0^t \left(\phi_0(t-\tau) I + \phi_1(t-\tau) A + \dots + \phi_{n-1}(t-\tau) A^{n-1}\right) B u(\tau)\, d\tau$$

$$= \int_0^t \phi_0(t-\tau) B u(\tau)\, d\tau + \int_0^t \phi_1(t-\tau) AB u(\tau)\, d\tau$$

$$+ \dots + \int_0^t \phi_{n-1}(t-\tau) A^{n-1} B u(\tau)\, d\tau$$

$$= \begin{bmatrix} B & AB & \dots & A^{n-1}B \end{bmatrix} \begin{bmatrix} \int_0^t \phi_0(t-\tau) u(\tau)\, d\tau \\ \int_0^t \phi_1(t-\tau) u(\tau)\, d\tau \\ \vdots \\ \int_0^t \phi_{n-1}(t-\tau) u(\tau)\, d\tau \end{bmatrix} \longrightarrow ⑪$$

$$\boxed{x(t) \in \text{img}\begin{bmatrix} B & AB & \dots & A^{n-1}B \end{bmatrix}}$$

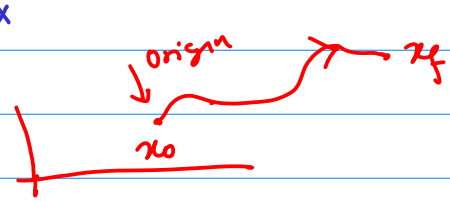
$$\dim\left(\text{img}\begin{bmatrix} B & AB & \dots & A^{n-1}B \end{bmatrix}\right) = n \Rightarrow \mathbb{R}^n \text{ is reachable.}$$

$$\underline{R_t \subseteq \text{img}\, \mathcal{C}_{AB}}$$

$$R_t = \text{img}\, \mathcal{C}_{AB}$$

$$\text{rank}\underbrace{\begin{bmatrix} B & AB & \dots & A^{n-1}B \end{bmatrix}}_{\text{controllability matrix}} = n \Leftrightarrow \text{System is Controllability}$$

Task: Prove the thm for contr.

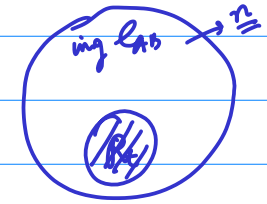


$$\mathcal{C}_{AB} := \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}$$

$$\mathcal{R}_t = \{ \xi \in \mathbb{R}^n \mid \exists\, u(t) \text{ that steers the system from } 0 \text{ to } \xi \text{ in finite time} \}$$

Reachable space

$$\boxed{\mathcal{R}_t \subseteq \operatorname{img} \mathcal{C}_{AB}} \longrightarrow \text{(A) } \checkmark$$



$$\square \quad W_t := \int_0^t e^{A\tau} B B^* e^{A^*\tau}\, d\tau$$

Controllability Gramian

$$\dim \operatorname{img} \mathcal{C}_{AB} = n \Updownarrow \mathcal{R}_t = \mathbb{R}^n$$

Thm: $\operatorname{img} \mathcal{C}_{AB} \subseteq \operatorname{img} W_t$

Given a subspace $\mathcal{X}$ we define orthogonal complement of $\mathcal{X}$ by

$$\mathcal{X}^\perp = \{ x \in \mathbb{R}^n \mid x^* y = 0 \ \forall\, y \in \mathcal{X} \}$$

e.g. $\mathcal{X} = \operatorname{img} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \subseteq \mathbb{R}^3$, $\mathcal{X}^\perp = \operatorname{img} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ $\quad v = \alpha \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$\downarrow$$

$$w = \beta_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \beta_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$v^* w = \alpha^* \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \left(\beta_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \beta_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = 0$$

$$\boxed{\dim(\mathcal{X}^\perp) = n - \dim(\mathcal{X})}$$

Proposition: Given any matrix P, it satisfies

$$(\operatorname{img} P)^\perp = \ker P^*$$

Lemma: $\operatorname{img} \mathcal{C}_{AB} \subseteq \operatorname{img} W_t \iff (\operatorname{img} \mathcal{C}_{AB})^{\perp} \supseteq (\operatorname{img} W_t)^{\perp}$

$$W_t = \int_0^t e^{A\tau} B B^* e^{A^*\tau} d\tau$$

$$W_t^* = \int_0^t e^{A\tau} B B^* e^{A^*\tau} d\tau = W_t$$

$$v \in (\operatorname{img} W_t)^{\perp} \Rightarrow v \in \ker W_t^* \Rightarrow v \in \ker W_t$$

$$\Downarrow$$

$$\left(\int_0^t e^{A\tau} B B^* e^{A^*\tau} d\tau\right) v = 0$$

$$\Rightarrow v^* \left(\int_0^t e^{A\tau} B B^* e^{A^*\tau} d\tau\right) v = 0$$

$$\Rightarrow \int_0^t v^* e^{A\tau} B B^* e^{A^*\tau} v \, d\tau = 0$$

Let $B^* e^{A^*\tau} v =: p(\tau)$

$p^*(\tau) = v^* e^{A\tau} B$

$$\Rightarrow \int_0^t p^*(\tau) p(\tau) \, d\tau = 0 \Rightarrow p(\tau) = 0 \quad \forall\ 0 \leq \tau \leq t$$

$$\Rightarrow p^*(\tau) = 0 \quad \forall\ 0 \leq \tau \leq t$$

$$\Downarrow$$

$$v^* e^{A\tau} B = 0 \quad \forall\ 0 \leq \tau \leq t \longrightarrow \text{(III)}$$

$\tau = 0$, $\quad v^* B = 0$

Diff (III) w.r.t. τ, $\quad v^* e^{A\tau} A B = 0 \quad$ For $\tau = 0$, $v^* A B = 0$

2nd diff (III) w.r.t τ, $\quad v^* e^{A\tau} A^2 B = 0 \quad$ For $\tau = 0$, $v^* A^2 B = 0$

$$\vdots$$

$$v^* \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix} = 0 \Rightarrow v \in (\operatorname{img} \mathcal{C}_{AB})^{\perp}$$

$$\boxed{\operatorname{img} \mathcal{C}_{AB} \subseteq \operatorname{img} W_t} \longrightarrow \text{(B)}$$

Lemma: $\quad \operatorname{img} W_t \subseteq R_t$

Select at any time t, $p \in \operatorname{img} W_t$

$\exists\, v$ s.t. $\quad W_t v = p$

Now define $\quad u(\tau) := \underline{\underline{B^* e^{A^*(t-\tau)} v}} \longrightarrow *$

Solution of $\dot{x} = Ax + Bu$, $x(0) = 0$ due to input $u(\tau)$ is

$$x(t) = \int_0^t \underline{e^{A(t-\tau)} B \underline{\underline{B^* e^{A^*(t-\tau)} v}}}\, d\tau \qquad \underbrace{\int_0^t u^*(\tau)\, u(\tau)\, d\tau}_{\text{Energy}}$$

Let $t - \tau = \xi$, $\quad d\tau = -d\xi$, $\quad \tau = 0,\ \xi = t$
$\tau = t,\ \xi = 0$

Status $\rightarrow$

$$\underline{\underline{x(t)}} = \int_t^0 e^{A\xi} B B^* e^{A^*\xi} v\, (-d\xi)$$

$$= \underbrace{\left(\int_0^t e^{A\xi} B B^* e^{A^*\xi}\, d\xi \right)}_{W_t} v$$

$$\boxed{x(t)} = \underbrace{W_t v}_{} = p \Rightarrow \quad x_t \in \operatorname{img} W_t$$

$\boxed{\operatorname{img} W_t \subseteq R_t} \longrightarrow$ (C)

$\boxed{R_t \subseteq \operatorname{img} \mathcal{C}_{AB}} \longrightarrow$ (A)

$\boxed{\operatorname{img} \mathcal{C}_{AB} \subseteq \operatorname{img} W_t} \longrightarrow$ (B)

$$\boxed{\begin{aligned} R_t &= \operatorname{img} W_t \\ &= \operatorname{img} \mathcal{C}_{AB} \end{aligned}} \quad *$$